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RESEARCH ARTICLE

A Lightweight PCB Defect Detection Algorithm Based on an Improved YOLOv11s

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ABSTRACT To address the limited accuracy of existing PCB surface defect detection methods, the challenge of balancing detection performance with lightweight design, and the need for real-time deployment in electronic manufacturing systems, we propose a lightweight PCB defect detection algorithm based on an improved YOLOv11s. First, a lightweight EfficientViT module is introduced into the backbone to replace the original backbone, reducing computational resource usage while enhancing inter-channel information exchange. Second, we integrate FasterNet partial convolution (PConv) with an efficient multi-scale attention mechanism (EMA), and design a C3k2_Faster_EMA convolution to replace standard convolutions in the neck module, further decreasing parameter count and strengthening fine-grained feature extraction. Finally, by combining Inner_IoU auxiliary bounding boxes with the dynamic focusing mechanism of Wise_IoU, we construct an Inner_Wise_IoU loss to replace the original CIoU loss, improving localization accuracy for small targets and bounding-box regression performance. Experiments show that the improved method achieves strong performance on the PKU-Market-PCB and DeepPCB datasets, attaining an mAP of 92.3%. Compared with the original YOLOv11s, parameters are reduced by 61.7%, computational complexity by 63.9%, and model size by 58.3%, down to 8.0MB, while reaching an inference speed of 227.54 FPS. To validate deployment feasibility on edge devices, we deploy the model on the LubanCat-5 platform. The results indicate that the quantized model size is 6.1MB, with an average frame rate of 23.094 FPS and an average inference time of 43.301ms; relative to the non-quantized version, the frame rate increases by 130% and inference time decreases by 56%. These results demonstrate that the proposed model achieves a favorable balance between lightweight design and detection accuracy, offering strong engineering value and practicality for modern electronic manufacturing systems.

INDEX TERMS PCB defect detection, YOLOv11s, EfficientViT, attention mechanism, loss function.

I. INTRODUCTION

With the rapid growth of the electronics manufacturing industry, printed circuit boards (PCBs), as fundamental components, play a pivotal role in determining the performance and reliability of electronic products [1]. However, PCB fabrication is relatively complex and can give rise to a wide variety of defects. Common defect types include missing holes, mouse-bite defects, open circuits, short circuits, spurious defects, and excess copper. Even such minute defects may cause equipment failures. Consequently, detecting tiny surface defects on PCBs has become a key issue [2], [3], [4]. Traditional approaches, such as manual visual inspection and automated optical inspection (AOI) [5], often suffer from low efficiency and are prone to missed detections and false alarms, making it difficult to satisfy the high-efficiency and high-precision requirements of modern production. Owing to its strong feature-learning capability, deep learning has

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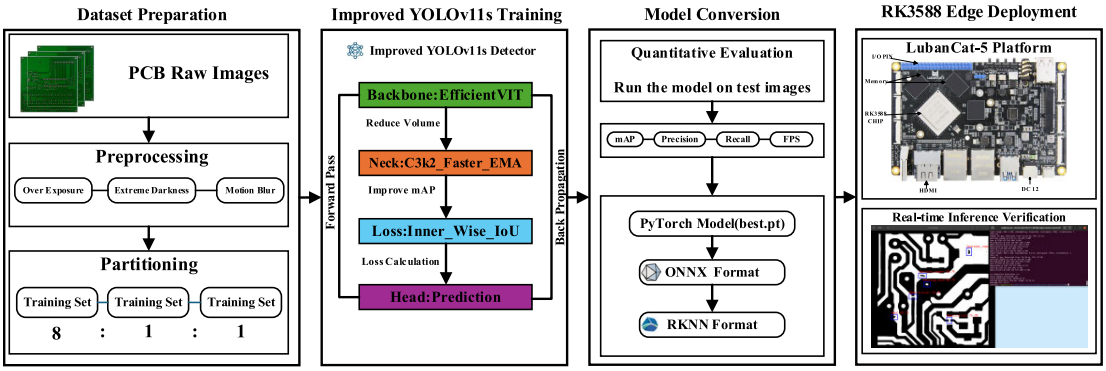


FIGURE 1. Workflow of the proposed PCB defect detection method.

markedly improved the automation level and accuracy of PCB defect detection, and has become an important direction in this field [6].

Current deep learning-based object detection algorithms are primarily divided into two major categories: one-stage algorithms and two-stage algorithms. Two-stage algorithms, primarily represented by CNN [7], R-CNN [8], and Fast R-CNN [9], perform candidate bounding box generation and region classification in hierarchical stages. Although they achieve higher accuracy, their detection speed is slow and cannot satisfy the real-time requirements of defect detection tasks. One-stage algorithms, represented by the YOLO and SSD series, directly generate target location information and class predictions, enabling end-to-end real-time detection, which makes them suitable for deployment in PCB surface defect detection scenarios.

Although many improvements have been proposed for PCB surface defect detection, existing models still exhibit limitations. Chen et al. [10] introduced a weighted bidirectional feature pyramid network together with the CA attention mechanism based on YOLOv5. While multi-scale feature fusion was improved, the complex cross-layer connections substantially increased memory-access overhead, leading to a noticeable drop in inference speed and limiting edge deployment. Wang et al. [11] improved YOLOv8s using dilated convolution (S-Conv) and the VFWD-CIoU loss. Although detection accuracy improved significantly, S-Conv may not guarantee complete feature extraction, and VFWD-CIoU weakens optimization on low-IoU samples, undermining the original efficiency and lightweight advantages and making highly efficient deployment difficult on resource-constrained hardware. Wang et al. [12] incorporated the SE (Squeeze-and-Excitation) attention mechanism into YOLOv8 and added a small-object detection scale to detect 4×4 defect targets. However, SE attention relies heavily on channel reweighting and may overlook spatial information, reducing robustness; moreover, the additional detection layer increases training time and parameter count, complicating deployment. In addition, Xu et al. [13] proposed a lightweight PCB surface defect detection framework, EYOLOv8, which reconstructs the backbone by introducing a lightweight RevCol convolutional network. Although parameters were greatly reduced, excessive model compression caused the loss of high-frequency feature information, preventing it from matching the latest YOLOv11 in terms of detection accuracy and real-time performance.

In summary, the above methods either rely on relatively outdated baselines, inflate parameter counts in pursuit of accuracy, or lose critical features due to over-lightweighting, and thus struggle to satisfy the dual requirements of high accuracy and real-time operation in electronic industrial systems. To address the above issues, this paper proposes a lightweight PCB defect detection algorithm based on an improved YOLOv11s, whose overall workflow is illustrated in Fig. 1. It employs a two-stage strategy of "first reducing scale, then compensating for accuracy." By replacing the lightweight backbone network, applying lightweight convolution and attention mechanisms, and employing a more efficient and accurate bounding box loss function, the approach simultaneously achieves lightweight model modification and enhanced detection accuracy.

II. IMPROVED YOLOv11s MODEL

YOLOv11 adopts a modular design that enables rapid optimization and customization. For instance, the C3k2 convolution module can be replaced or enhanced by fusing other modules as needed to suit different tasks; moreover, its flexible architecture facilitates the integration of attention mechanisms to improve detection accuracy. As a lightweight variant in the YOLOv11 family, YOLOv11s features fewer parameters while maintaining strong accuracy, allowing efficient operation on embedded devices or in computation-limited environments—an advantage well aligned with real-time PCB defect detection. Accordingly, this paper improves YOLOv11s in three aspects, and the modified architecture is shown in Fig. 2.

1) A sandwich-layout EfficientViT (Efficient Vision Transformers) module is introduced to replace the original backbone, improving memory efficiency, strengthening inter-channel information exchange, and reducing computational resource consumption.

TABLE 1. Correspondence between EfficientViT output features and neck inputs.

Stage	Operator	Output Stride	Output(H×W×C)	Export to Neck
Stem	Overlap PatchEmbed	4	160×160×32	No
Stage1	EfficientViT Blocks × L1	8	80×80×64	P3
Stage2	Subsample + EfficientViT Blocks × L2	16	40×40×128	P4
Stage3	Subsample + EfficientViT Blocks × L3	32	20×20×256	P5

TABLE 2. Stage-wise complexity statistics of the EfficientViT backbone network.

Component	Output (H×W×C)	Params/M	FLOPs/G
Stem(Overlap PatchEmbed)	160×160×32	0.23	0.72
Stage 1 (Blocks × L1)	80×80×64	0.51	1.22
Subsample (S1→S2)	40×40×128	0.07	0.29
Stage 2 (Blocks × L2)	40×40×128	0.87	2.12
Subsample (S2→S3)	20×20×256	0.13	0.41
Stage 3 (Blocks × L3)	20×20×256	1.09	2.74
Total	—	2.90	7.50

2) A C3k2_Faster_EMA module is designed to replace C3k2 in the neck, reducing parameters while maintaining recognition accuracy.

3) By combining the auxiliary bounding-box loss Inner_IoU with the dynamic focusing mechanism of Wise_IoU, an Inner_Wise_IoU loss is constructed to replace the original CIoU loss, further improving recognition accuracy and bounding-box regression performance.

A. BACKBONE NETWORK IMPROVEMENT

For PCB surface defect detection, the original YOLOv11 backbone incurs relatively high computational complexity, making it difficult to satisfy deployment requirements under resource constraints. Therefore, this paper adopts the lightweight and efficient EfficientViT as the backbone feature-extraction module, as illustrated in Fig. 3. EfficientViT is built upon the EfficientViT Block, where each block comprises a sandwich layout and cascaded group attention (Cascaded Group Attention, CGA). CGA partitions features into different groups and performs parameter reallocation, thereby alleviating the memory-access overhead introduced by attention computation. In addition, EfficientViT Blocks insert a memory-efficient multi-head self-attention (MHSA) mechanism between feed-forward network (FFN) layers, which improves memory efficiency while enhancing inter-channel information exchange. Meanwhile, a DWConv layer is placed before each FFN layer to facilitate information interaction among local tokens and to introduce inductive biases. Overall, this design substantially reduces computational complexity and memory demand, while also mitigating redundant computation in multi-head attention.

To maintain the stability of the detection head, this paper generates three feature maps with strides {8, 16, 32} in the EfficientViT network, which are sequentially output as {P3, P4, P5} to the YOLO Neck; the corresponding relationships are shown in Table 1. Specifically, these three feature paths are fed into the original YOLO neck network for feature fusion in exactly the same way as the baseline backbone outputs, thereby ensuring fairness in the comparative experiments while enabling EfficientViT to serve as a plug-and-play backbone replacement scheme. Meanwhile, the Subsample layer downsamples the spatial resolution by a factor of 2 between stages while simultaneously increasing the channel width. In the implementation of this paper, this layer consists of a stride-2 depthwise 3×3 convolution and a 1×1 pointwise convolution for adjusting the channel dimension. Each convolution layer is followed by Batch Normalization and SiLU activation functions. Compared with the sampling approach using standard 3×3 convolution, this design achieves efficient downsampling at significantly lower computational cost.

To analyze the complexity variation of the proposed backbone network, this paper presents stage-wise computational complexity statistics in Table 2. At an input resolution of 640×640, the parameter count and computational complexity of this backbone network are 2.90M and 7.50G, respectively. It is noteworthy that the two Subsample operators introduce only minor additional overhead; the computational load is primarily concentrated in Stage 2 and Stage 3, mainly due to the increase in channel width, indicating that the design achieves an effective trade-off between representational capacity and computational cost.

B. C3k2 MODULE IMPROVEMENT

YOLOv11 employs the C3k2 module for feature extraction. Unlike the C2f module in YOLOv8, C3k2 reduces

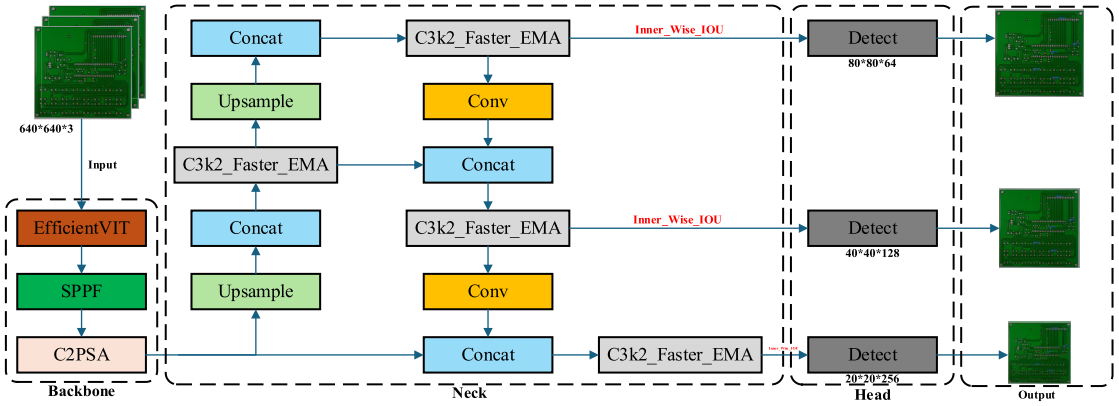


FIGURE 2. Improved YOLOv11s network architecture.

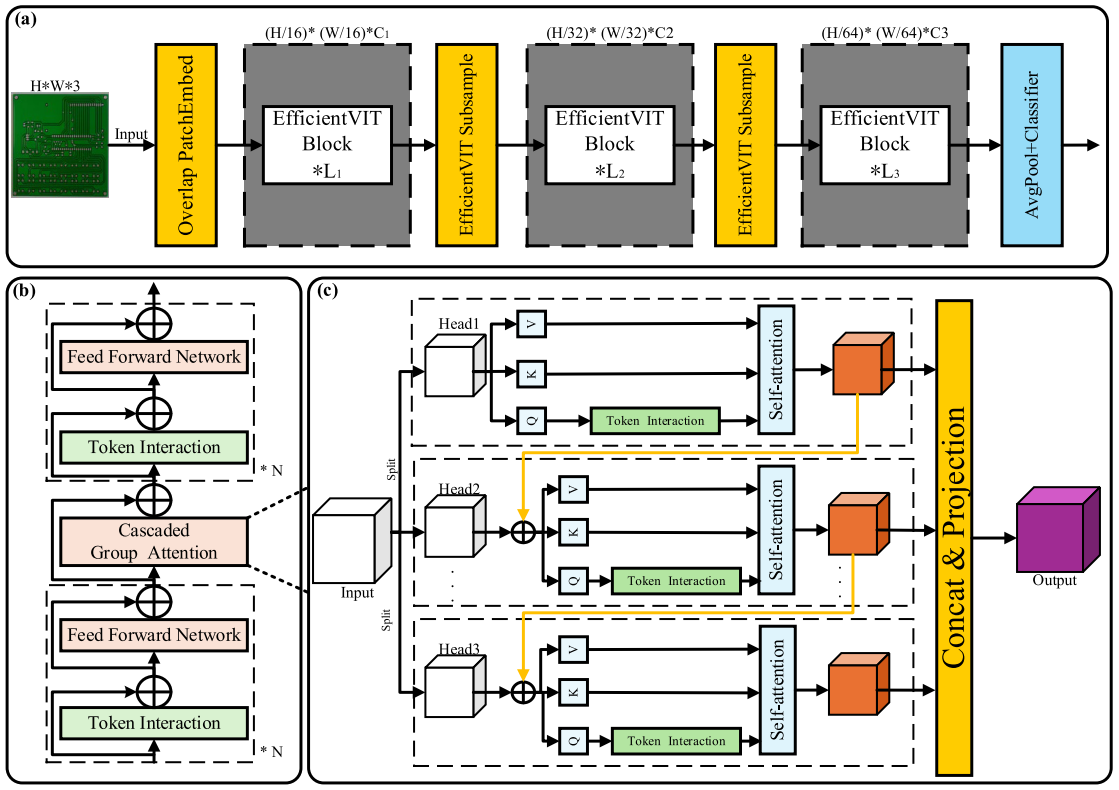


FIGURE 3. EfficientViT architecture. (a) EfficientViT; (b) Sandwich Layout; (c) Cascaded group attention.

computational complexity and parameter count by optimizing kernel convolutions and simplifying the design. However, in deeper networks, extensive use of standard C3k2 convolutions still incurs substantial computational overhead and memory consumption, leading to slower detection speed and making deployment more difficult, and thus it is less suitable for PCB defect detection. To reduce model complexity while maintaining recognition accuracy, the C3k2 module must be improved. Accordingly, we introduce FasterNet's Partial Convolution (PConv) [14] together with a convolutional attention module, Efficient Multi-scale Attention (EMA) [15].

FasterNet uses partial convolution to handle irregular holes in images, replacing standard convolution layers so as to ignore invalid pixels and perform conditional computation on valid pixels. PConv consists of two components: Masked and Re-normalized Convolution and a Mask-Update Step. Its structure is shown in Fig. 4. The mask-update step examines the number of valid pixels within a local window to progressively mark output positions.

In a partial convolution layer, each convolution operation performs a weighted sum only over valid pixels (i.e., regions where the mask value equals 1). Let the input feature map be $X \in R^{h\times w\times C_P}$, the corresponding binary mask be

$M \in \{0, 1\}^{h \times w}$, and the convolution kernel be $W \in R^{k \times k \times C_p}$, then the output of PConv can be expressed as:

$$Y_{i,j} = \begin{cases} \dfrac{1}{\sum_{u,v} M_{i+u,j+v} + \varepsilon} \sum_{u,v} W_{u,v} \left(X_{i+u,j+V} \cdot M_{i+u,j+v} \right) + b, \\ \qquad if \ \sum_{u,v} M_{i+u,j+v} \succ 0 \\ 0, \qquad otherwise \end{cases} \tag{1}$$

Here, ε denotes a very small constant, and b is the bias term. After each convolution operation, the mask is updated as:

$$M'_{i,j} = \begin{cases} 1, & f \sum_{u,v} M_{i+u,j+v} \succ 0, \\ 0, & otherwise \end{cases} \tag{2}$$

Through the above convolution operation, it is ensured that the missing regions in the input features will not cause erroneous propagation to the output, thereby realizing content-based adaptive filling.

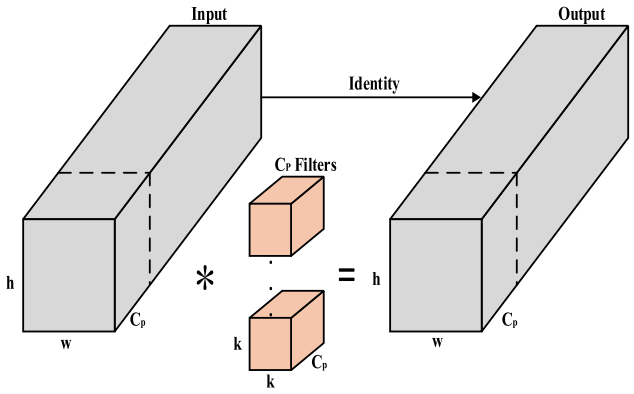


FIGURE 4. PConv architecture.

EMA is an efficient multi-scale attention module that enhances the feature representation capability of convolutional neural networks through cross-spatial learning. Compared with other attention mechanisms, it achieves superior performance while maintaining a small number of parameters. Its architecture is shown in Fig. 5. EMA employs three parallel sub-branches to capture attention-weight descriptors for grouped feature maps. Two sub-branches lie in the 1×1 branch, where spatial information is encoded via horizontal and vertical one-dimensional average pooling, while the third sub-branch lies in the 3×3 branch. The outputs of these branches are combined through matrix dot products to produce the first spatial attention map. In a similar manner, two-dimensional global average pooling is used in the 3×3 branch to aggregate global spatial details, yielding a second spatial attention map. Finally, for each group, the output feature map is computed by aggregating the weights from the two spatial attention maps and then applying a Sigmoid function; the final EMA output has the same spatial dimensions as the input.

In this model, the convolution in the Bottleneck structure of the C3k2 module is replaced with Pconv to obtain the C2f_Faster module, and then the EMA attention mechanism is added to the Bottleneck structure to obtain the C3k2_Faster_EMA module. The specific structure is shown in Fig. 6. Through the above improvements, the number of model parameters can be reduced, helping the model strengthen feature representation at a finer-grained level and improve the reconstruction ability of local details. At the same time, applying the attention mechanism at different stages of feature extraction helps to learn feature information of different levels at multiple hierarchies, thereby better capturing global and local features in images, effectively addressing the problem of accuracy degradation after model lightweighting, and improving the robustness and convergence stability of the model.

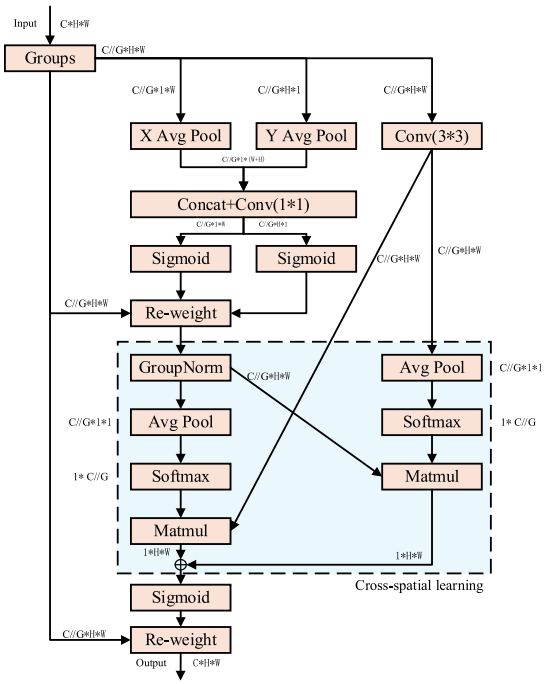


FIGURE 5. EMA architecture.

C. LOSS FUNCTION IMPROVEMENT

In object detection, the loss function is a critical component [16], and a well-designed bounding-box loss can substantially enhance model performance. YOLOv11 adopts CIoU as its loss and follows an anchor-free paradigm. By leveraging factors such as the overlap area between bounding boxes and the distance between their center points to guide prediction, the model can better accommodate targets of varying scales and shapes. However, in small-object detection, bounding-box regression still faces two issues: **(i)** even slight localization errors can significantly affect detection outcomes; **(ii)** sample-quality imbalance during training, where gradients from low-quality samples interfere

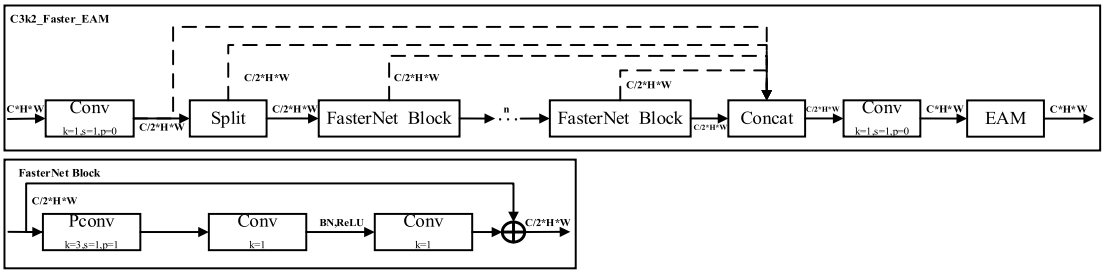


FIGURE 6. C3k2_Faster_EMA architecture.

with learning from high-quality samples, thereby degrading overall training effectiveness. To address these problems, this paper introduces the Inner_IoU Loss and Wise_IoU Loss.

To improve bounding-box localization accuracy, we incorporate an auxiliary bounding-box loss, Inner_IoU, defined as:

$$b_l^{\mathrm{gt}} = x_c^{\mathrm{gt}} - \frac{w^{gt} \times ratio}{2}, b_r^{gt} = x_c^{gt} + \frac{w^{gt} \times ratio}{2} \tag{3}$$

$$b_t^{gt} = y_c^{gt} - \frac{h^{gt} \times ratio}{2}, b_b^{gt} = y_c^{gt} + \frac{h^{gt} \times ratio}{2} \tag{4}$$

$$b_l = x_c - \frac{w \times ratio}{2}, b_r = x_c + \frac{w \times ratio}{2} \tag{5}$$

$$b_t = y_c - \frac{h \times ratio}{2}, b_b = y_c + \frac{h \times ratio}{2} \tag{6}$$

$$\begin{aligned} \mathrm{inter} = &\left(\min\left(b_r^{gt}, b_r\right) - \max\left(b_l^{gt}, b_l\right)\right) \\ &\times \left(\min\left(b_b^{gt}, b_b\right) - \max\left(b_t^{gt}, b_t\right)\right) \end{aligned} \tag{7}$$

$$union = \left(w^{gt} \times h^{gt}\right) \times (ratio)^2 + (w \times h) \times (ratio)^2 - \mathrm{inter} \tag{8}$$

$$IoU^{inner} = \frac{\mathrm{inter}}{union} \tag{9}$$

In the formula: b^{gt} represents the GT box; b represents the anchor box; $\left(x_c^{gt}, y_c^{gt}\right)$ represents the center-point coordinates of the GT box; (x_c, y_c) represents the center-point coordinates of the anchor box; w and h represent the width and height, respectively; the variable *ratio* represents a scale factor, with a value range of [0.5,1.5]. Subsequently, by adjusting the value of ratio, a bounding-box size with higher localization accuracy is found, thereby improving recognition accuracy.

To address the issue of imbalanced training sample quality, Wise_IoU Loss is introduced, which comprises three versions: Wise_IoU_v1, Wise_IoU_v2, and Wise_IoU_v3, defined as:

$$R_{WIoU} = exp\left(\frac{\left(x - x_{gt}\right)^2 + \left(y - y_{gt}\right)^2}{\left(W_g^2 + H_g^2\right)^*}\right) \tag{10}$$

$$L_{WIoUv1} = R_{WIoU} L_{IoU} \tag{11}$$

$$L_{WIoUv2} = \gamma L_{WIoUv1} \tag{12}$$

$$L_{WIoUv3} = \rho L_{WIoUv1}, \rho = \frac{\beta}{\delta \alpha^{\beta - \delta}} \tag{13}$$

In the formula: $\gamma = L_{IoU}^{\gamma *} \in [0, 1]$ denotes the gradient gain factor; when $\beta = \mathrm{C}$, the gradient gain reaches its maximum.

Because L_{IoU} changes dynamically, the criterion for categorizing anchor-box quality is also dynamic. Leveraging a dynamic non-monotonic focusing mechanism, Wise_IoU evaluates anchor-box quality and accordingly adjusts the loss weights, thereby allocating gradient gains in a manner that best matches the current training state. This encourages the model to focus more on learning from medium-quality samples, while preventing harmful gradients from low-quality samples from disrupting training, thus improving detection accuracy.

In the proposed model, the original bounding box loss function is replaced with Inner_IoU, where the bounding box size is adjusted via the *ratio* parameter to accelerate model convergence. After subsequent comparative experiments on the three versions of Wise_IoU, Wise_IoU_v3 was found to deliver the best improvement in model accuracy. Therefore, this paper combines the more localization-sensitive Inner_IoU with the dynamic focusing mechanism of Wise_IoU_v3 to construct the Inner_Wise_IoU loss function, which improves bounding box localization accuracy, optimizes sample quality distribution, and further enhances model detection precision. To ensure consistency in the optimization objective, the internal similarity term is rewritten in loss form: $L_{inner} = 1 - IoU_{inner}$.The final bounding box regression loss is defined as:

$$L_{Inner_Wise_IoU} = L_{WIoU_v3} + \lambda\left(1 - IoU_{inner}\right) \tag{14}$$

In the formula: λ represents the contribution weight controlling the internal box constraint term, by encouraging the predicted bounding box to achieve better overlap in the internal region, the localization accuracy of small defect targets is improved.

III. EXPERIMENTAL ANALYSIS

A. DATASETS

To evaluate the detection performance of the proposed method, experiments were conducted on two public datasets: PKU-Market-PCB [17] and DeepPCB [18]. Both datasets were randomly split into training, validation, and test sets at a ratio of 8:1:1.

TABLE 3. PKU-Market-PCB dataset information.

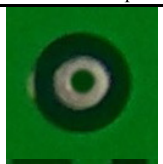
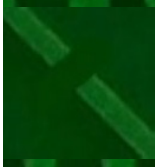
Defect category	Defect example	Defect count
Missing hole		497
Mouse bite		492
Open circuit		482
Short		491
Spur		488
Spurious copper		503

TABLE 4. DeepPCB dataset information.

Defect category	Defect example	Defect count
Open circuit		1702
Short		1317
Mouse bite		1748
Spur		1445
Spurious copper		1320
Missing hole		1320

1) PKU-MARKET-PCB DATASET

The PKU-Market-PCB dataset was released by Peking University's Open Laboratory of Intelligent Robotics and contains six defect categories: Missing hole, Mouse bite, Open circuit, Short, Spur, and Spurious copper. It includes a total of 693 defect images, and detailed information is provided in Table 3.

2) DeepPCB DATASET

The DeepPCB dataset was released by Shanghai Jiao Tong University and includes six defect types: Open circuit, Short, Mouse bite, Spur, Spurious copper, and Missing hole, with a total of 1,500 images. Unlike PKU-Market-PCB, each image in DeepPCB contains multiple defects of different categories, reflecting PCB defects in real-world manufacturing environments. Moreover, all images were acquired via CCD (Charge-Coupled Device) scanning and thus exhibit high resolution and clarity. Detailed information is provided in Table 4.

TABLE 5. Experimental parameters.

Parameter name	Parameter value
Image size	640*640
Initial learning rate	0.01
Final learning rate	0.001
Optimizer	SGD（0.937）
Cosine annealing factor	0.1
Weight decay	0.0005
Batch size	16
Training epochs	300

B. EXPERIMENTAL ENVIRONMENT AND PARAMETER SETTINGS

To ensure the effectiveness of model training, all experiments in this paper were conducted under the same hardware environment, using the Windows 11 operating system and an NVIDIA GTX 5060Ti graphics card (16 GB), with the experimental environment consisting of Python 3.9, CUDA 12.9, and the PyTorch 2.9 deep learning framework.

TABLE 6. Ablation experiment results.

YOLOv11s	Backbone	C3k2	Loss	P/%	R/%	mAP/%	Params/106	FLOPs/G	ModelSize/MB
✓				93.6±0.70	88.5±0.92	91.6±0.38	9.4	21.6	19.2
✓	✓			92.5±0.78	84.9±1.05	89.8±0.42	3.7	8.0	8.3
✓	✓	✓		92.2±0.71	87.9±0.98	90.5±0.35	3.6	7.8	8.0
✓	✓	✓	✓	93.0±0.62	88.9±0.85	92.3±0.31	3.6	7.8	8.0

TABLE 7. Comparative results of backbone network replacement.

YOLOv11s	P/%	R/%	mAP/%	Params/106	FLOPs/G	ModelSize/MB
+MobileNetV4	88.2±0.72	79.0±1.05	86.0±0.45	9.5	27.8	19.5
+FasterNet	92.1±0.68	91.1±0.92	93.0±0.38	7.5	15.6	15.4
+StarNet	92.5±0.75	84.2±1.08	88.6±0.42	5.5	11.4	11.2
+EfficientFormerV2	92.2±0.70	90.7±0.95	93.2±0.35	8.4	16.5	68.5
+RepViT	92.3±0.65	89.1±0.88	92.2±0.40	10.1	24.1	20.9
+EfficientVIT(Ours)	**92.5±0.62**	**84.9±0.85**	**89.8±0.31**	**3.6**	**7.8**	**8.3**

TABLE 8. Comparative results of attention mechanisms.

YOLOv11s_C3k2_Faster	P/%	R/%	mAP/%	Params/10^6	FLOPs/G	ModelSize/MB
+SE	90.8±0.71	86.3±0.98	89.5±0.39	3.8	8.2	8.4
+ECA	91.3±0.69	86.7±0.95	89.9±0.36	3.7	8.0	8.2
+CBAM	91.6±0.73	87.0±0.92	90.0±0.37	4.0	8.5	8.9
+CoordAtt	91.8±0.67	87.2±0.90	90.2±0.34	3.9	8.3	8.6
+EMA(Ours)	**92.2**±0.63	**87.9**±0.87	**90.5**±0.32	**3.6**	**7.8**	**8.0**

No pre-trained weights were loaded, and the specific experimental parameters are shown in Table 5. All subsequent experimental results are reported as the mean ± standard deviation from five independent training runs to verify the statistical robustness of the results.

C. EVALUATION METRICS

The experiments in this paper use precision (P), recall (R), mean average precision (mAP), model size (Modelsize), number of model parameters (Params), floating-point operations (FLOPs), and frames per second (FPS) as evaluation metrics.

$$P = \frac{TP}{TP + FP} \times 100\% \tag{15}$$

$$R = \frac{TP}{TP + FN} \times 100\% \tag{16}$$

$$AP = \int_0^1 P(R)dr \tag{17}$$

$$mAP = \frac{1}{M}\sum_{i=1}^{M} AP(i) \times 100\% \tag{18}$$

In the formula, TP is the true positive, indicating the number of samples that the model correctly predicts as the positive class; FN is the false negative, indicating the number of samples that the model incorrectly predicts as the negative class; FP is the false positive, indicating the number of samples that the model incorrectly predicts as the positive class; AP is the average precision; M is the number of detected samples; N is the total number of images processed by the model within time t; mAP reflects the overall accuracy of the model, the larger its value, the higher the recognition accuracy of the model.

TABLE 9. Loss-function comparison results.

Loss-Function Comparison		Inner-Parameter Comparison	
Loss function	mAP/%	*ratio*	mAP/%
CIoU	90.5	0.70	91.6
WIoUv1-Inner-ShapeIoU	91.1	0.85	91.8
WIoUv2-Inner-ShapeIoU	91.0	0.90	92.3
WIoUv3-Inner-MPDIoU	90.8	0.95	91.4
WIoUv3-Inner-ShapeIoU	91.9	1.05	92.0

D. ABLATION EXPERIMENTS

To verify the effectiveness of each improvement, ablation experiments were used for comparative analysis, and the ablation results are shown in Table 6. Based on the YOLOv11s model, the backbone trunk of the backbone network was

TABLE 10. Comparison results with different algorithms.

Model	Params/10^6	FLOPs/G	ModelSize/MB	mAP/%	FPS
YOLOv5s	7.8	19.0	16.0	90.8±0.55	169.28±5.2
YOLOv8s	9.8	23.6	20.0	87.1±0.68	183.20±6.5
YOLOv9s	6.3	22.7	13.3	91.8±0.46	159.76±4.8
rtdetr-x	67.3	232.4	270.3	92.2±0.40	45.33±2.3
YOLOv11s	9.4	21.6	19.2	91.6±0.42	204.65±7.1
YOLOv11s-MobileNetV4	5.4	21.1	11.3	90.5±0.57	95.80±4.9
Ours	3.6	7.8	8.0	92.3±0.31	227.54±7.8

TABLE 11. Comparison of generalization experiment results.

Model	Params/10^6	FLOPs/G	ModelSize/MB	mAP/%	FPS
YOLOv5s	9.6	25.5	24.2	76.7±0.45	168.45±6.8
YOLOv8s	11.7	28.1	29.2	78.1±0.48	142.89±5.9
YOLOv9s	7.9	25.3	19.0	81.6±0.40	149.32±5.7
rtdetr-x	68.5	235.7	268.9	78.5±0.37	42.18±2.1
YOLOv11s	11.2	26.5	27.0	82.4±0.39	200.17±8.5
YOLOv11s-MobileNetV4	7.8	19.3	17.6	76.2±0.52	112.45±5.3
Ours	6.7	13.5	14.2	85.0±0.34	320.72±12.6

replaced with a lightweight EfficientVIT module. The model's parameter count, computational cost, and model size decreased by 60.6%, 63.0%, and 56.8%, respectively, fully indicating that this module plays a major role in lightweighting the original model. However, it is accompanied by a decrease in accuracy, and the mAP decreased by 1.8%. To further reduce the model parameters, the convolution module of the neck network was replaced with the C3k2_Faster_EMA module, and the overall parameter count of the model decreased, while the mAP increased by 0.7%, achieving recovery of accuracy loss while continuously compressing the model scale. To further enhance model accuracy, Inner_Wise_IOU was adopted to replace the original CIoU loss function. The final improved model achieved reductions of 61.7%, 63.9%, and 58.3% in parameter count, computational complexity, and model size, respectively, compared with the original model, while increasing mAP by 0.7%, fully validating that this loss function provides gains in bounding box regression accuracy without introducing additional computational overhead.

E. COMPARATIVE EXPERIMENTS

To comprehensively validate the effectiveness and superior performance of the proposed model in PCB defect detection tasks, this section conducts comparative analysis across five dimensions: First, experiments comparing backbone networks, attention mechanisms, and loss functions demonstrate the model's Pareto optimality. Second, experiments contrasting different algorithms highlight the model's superior performance. Finally, experiments evaluating generalization capabilities validate the model's adaptability. The models involved in the aforementioned experiments employed the same training environment and parameter configurations.

1) LIGHTWEIGHT BACKBONE NETWORK COMPARATIVE EXPERIMENTS

The lightweighting process of the proposed model is primarily achieved by replacing the lightweight backbone network. To verify its effectiveness in reducing model size and parameter count, EfficientViT is compared with lightweight backbones such as MobileNetV4 [19], FasterNet [20], StarNet [21], EfficientFormerV2 [22], and RepViT [23] based on the YOLOv11s model, and the results are shown in Table 7. Considering the resource constraints of edge devices, the model size must be less than 10 MB [24]. Except for EfficientViT, the other lightweight backbones generally exhibit larger model sizes. If these backbones are adopted, subsequently introducing attention mechanisms or loss functions in the Neck/Head will rapidly reach the resource upper limit, thereby limiting the adjustable space available for accuracy improvement. In contrast, EfficientViT achieves a stronger compression effect among all candidate backbones, providing more abundant redundancy space for subsequent module accuracy compensation, which enables subsequent mAP improvements to be realized under strict model size constraints and aligns with the overall optimization objective for resource-constrained scenarios.

2) ATTENTION MECHANISM COMPARATIVE EXPERIMENTS

Based on the EfficientViT lightweight backbone, this paper redesigns the structure of the C3k2 module in the neck to further reduce model size and improve model accuracy.

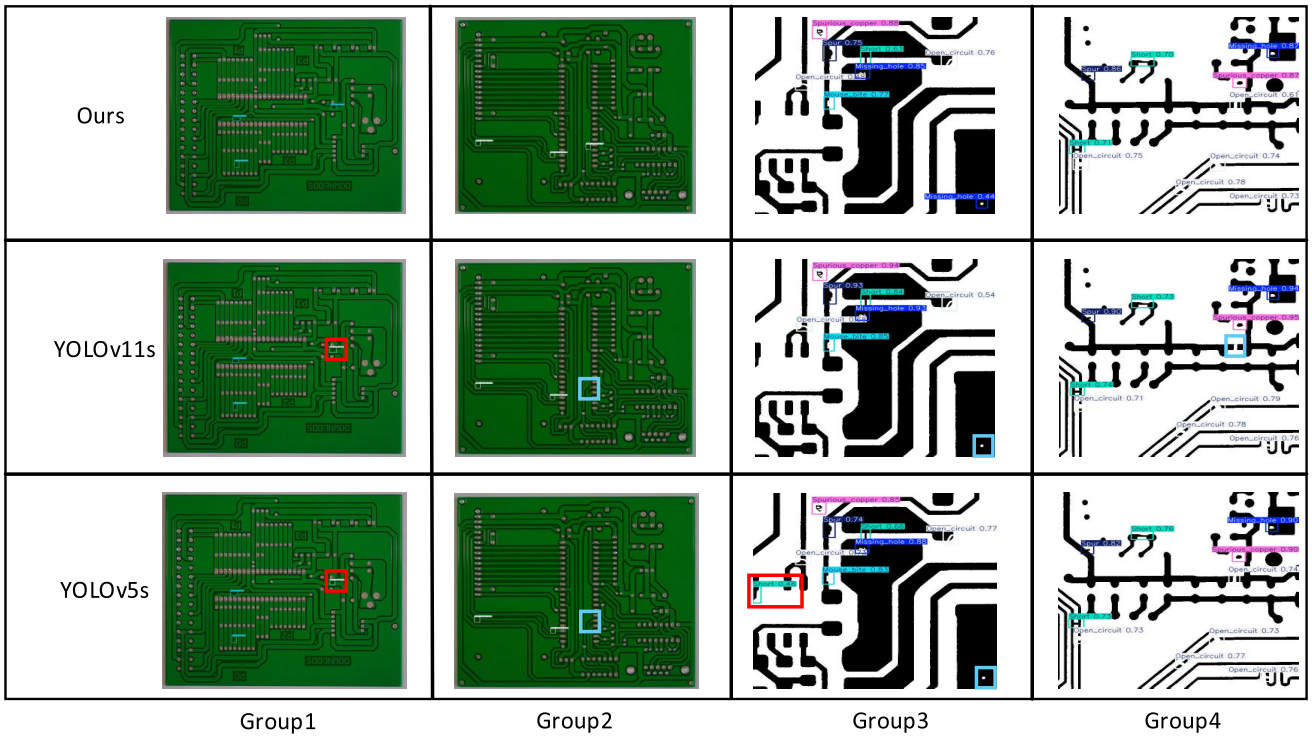


FIGURE 7. Qualitative comparison of detection results.

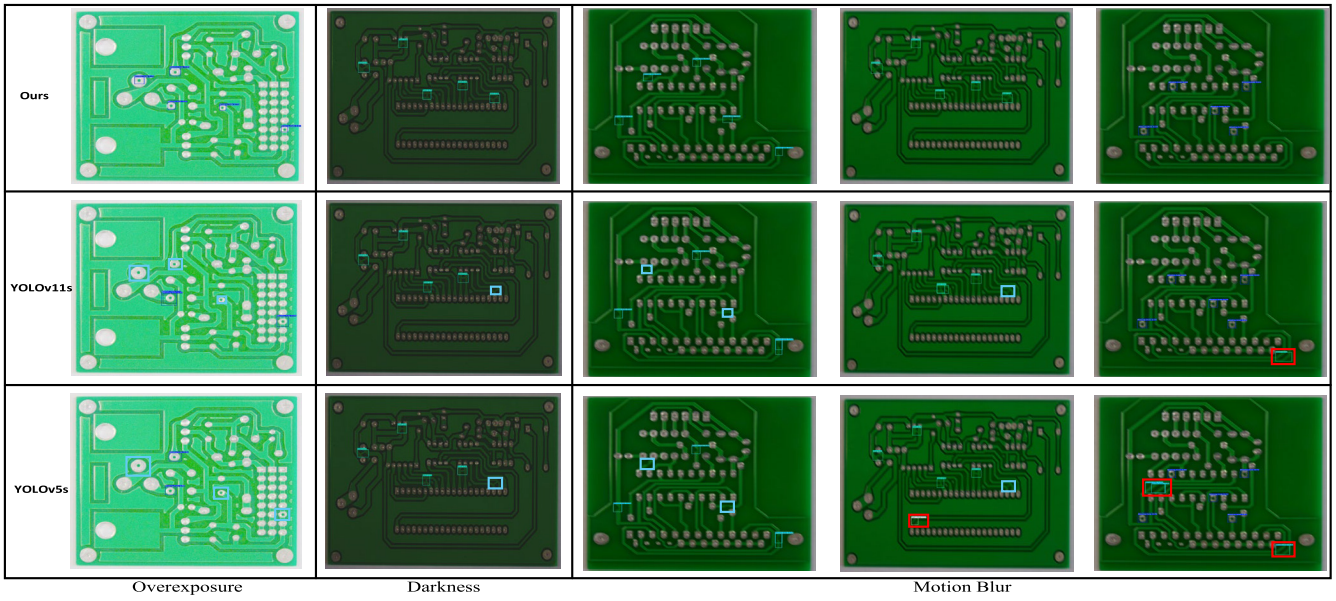


FIGURE 8. Detection results under different operating conditions.

To quantitatively evaluate the impact of different attention mechanisms on the improved C3k2 module, EMA is compared with attention mechanisms such as SE [25], ECA [26], CBAM [27], and CoordAtt [28], with results shown in Table 8. The results indicate that EMA achieves the best overall performance, improving by 1.0%, 0.6%, 0.5%, and 0.3% compared with SE, ECA, CBAM, and CoordAtt, respectively, while simultaneously enhancing both P and R. Meanwhile, EMA delivers superior complexity benefits while improving accuracy, incurring the smallest additional overhead and exhibiting better synergistic compression effects with the PConv module. In contrast, although CBAM provides certain accuracy gains, it significantly increases the number of parameters and computational load. In summary,

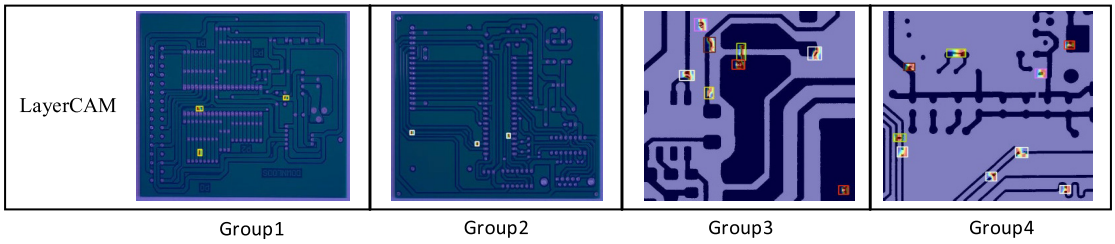


FIGURE 9. Heatmap visualization results.

TABLE 12. Hardware detailed information.

Device	Parameter information
CPU	Quad-core ARM Cortex-A55, Quad-coreARM Cortex-A76,Neon and FPU
GPU	ARMMali-G610 MP4, OpenGLES 1.1/2.0/3.2, Vulkan 1.2, openCL 2.2
AI	6TOPS
NPU	Support INT4/INT8/INT16/FP16
Memory/Storage	LPDDR4X-4 GB/eMMC-32 GB
Dimensions	190mm x125mm
On-board OS	Ubuntu-20.04.6

EMA best aligns with the attention enhancement scheme for the improved C3k2 module in this paper and is adopted to construct the final C3k2_Faster_EMA structure.

3) LOSS-FUNCTION COMPARISON EXPERIMENTS

To verify the effectiveness of the loss function improvement on the increase of mAP, loss function comparison experiments and Inner parameter comparison experiments were conducted respectively, and the results are shown in Table 9. Compared with WIoU v1 and v2, WIoUv3 has the best effect, and the mAP increases by 0.8% and 0.9% respectively, while the fused ShapeIoU increases mAP by 1.1% compared with MPDIoU [29]. In the Inner parameter comparison experiment, when ratio is set to 0.90, the mAP is the highest, reaching 92.3%.

4) COMPARISON EXPERIMENTS OF DIFFERENT ALGORITHMS

To verify the advantages of the proposed model over other models, the model was compared with algorithms such as YOLOv5s, YOLOv8s, YOLOv9s, rtdetr-x, YOLOv11s, and YOLOv11s-MobileNetV4. The experimental results are shown in Table 10. The proposed model shows high superiority in both lightweight metrics and detection accuracy.

From the experimental data, it can be obtained that the proposed model performs best in both lightweighting and detection accuracy. Compared with the classic models of the YOLO series, the mAP is increased by 1.5%, 5.2%, 0.5%, and 0.7% respectively, and the parameter count and computational cost are the lowest; compared with YOLOv11s improved by the lightweight MobileNetV4 network, the parameter count and computational cost are reduced by 33.33% and 63.03% respectively, mAP is increased by 2.0%, and FPS is increased by 137.50%; compared with the large rt-detr-x model, although the mAP increase is small, the parameter count and computational cost are reduced by 94.65% and 96.64% respectively, and FPS is increased by 402.20%. The results show that while ensuring extreme lightweighting, the proposed model also greatly improves inference speed, and better balances model lightweighting and detection accuracy.

TABLE 13. Model size.

Model	Model size/MB		
	Pytorch model	ONNX model	RKNN model
Ours	8	15.12	10.10
YOLOv11s	27	52.97	20.88

5) GENERALIZATION PERFORMANCE COMPARISON EXPERIMENTS

To verify the generalization ability of the improved model, this paper uses the more challenging DsPCBSD+ dataset [30] for comparative experiments. This dataset was produced and made public by South China Agricultural University. On the basis of four main defect categories—copper scraps, copper missing, conductor scratches, and foreign objects—it is further subdivided into nine defect types: short circuit, burr, virtual copper, open circuit, mouse bite, broken hole, conductor scratches, conductor foreign objects, and substrate foreign objects. It contains a total of 10,259 defect images and 20,276 manually annotated defect targets, and is randomly divided

TABLE 14. Model quantization and performance evaluation results.

Model	Model Quantization Algorithm			Model size/MB	Average fps/s	Average inference time/ms	mAP/%
	Linear quantization	Asymmetric quantization	No quantization				
Ours	✓			6.10	21.289	46.972	91.9
		✓		**6.10**	**23.094**	**43.301**	**92.3**
			✓	10.10	10.050	99.501	92.1
YOLOv11s	✓			11.12	13.393	74.665	91.3
		✓		11.12	14.634	68.336	91.8
			✓	20.88	6.584	151.890	91.5

into training, validation, and test sets according to 8:1:1. Different from the PKU-Market-PCB and DeepPCB datasets, the PCBs in this dataset are miniaturized, and the corresponding small image details increase. The defect sizes are overall small and easy to be confused with the background. In addition, defect categories such as conductor scratches, conductor foreign objects, and substrate foreign objects have large differences in color, size, and shape within the class, with extremely strong intra-class variability. The experimental parameter settings are consistent with this paper, and the results are shown in Table 11. From the data in the table, it can be seen that compared with other models, the proposed model has better comprehensive performance on the DsPCBSD+ dataset, fully proving its superior generalization performance.

F. VISUAL ANALYSIS

In order to intuitively compare the detection effects of the model proposed in this paper and the original model, some images were randomly selected from the test set for testing, and the results are shown in Fig. 7. At the same time, in the actual PCB production line, factors such as external lighting and the high-speed movement of the conveyor belt will affect the model's recognition accuracy. Therefore, this paper simulated three working conditions, namely over exposure, extreme darkness, and motion blur, and the results are shown in Fig. 8. Among them, the red box indicates false detection targets, and the blue box indicates missed detection targets.

From the figure analysis, it can be seen that the model proposed in this paper has improved detection accuracy compared with the original models such as YOLOv11s and YOLOv5s. Specifically, the proposed model can detect targets that the original models cannot detect, effectively reducing the missed detection rate, and the YOLOv11s and YOLOv5s models have multiple false detection targets in Group1-5, which indicates that the proposed model has stronger adaptability and robustness in complex industrial scenarios. Especially when dealing with tiny targets, its performance advantage is more obvious.

In order to further reflect the feature attention mechanism of the proposed model in the PCB defect detection task, this study visualized the feature maps output by the model in the form of heat maps based on the LayerCAM [31] algorithm. The results are shown in Fig. 9, where the red region indicates high attention of the model and the blue region indicates low attention.

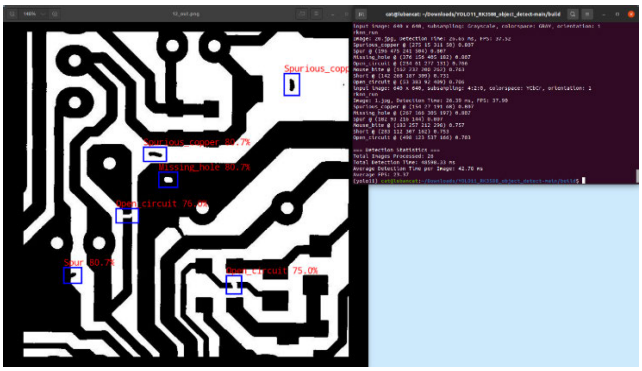


FIGURE 10. Schematic diagram of the RK3588 detection process.

G. EDGE-DEVICE DEPLOYMENT VERIFICATION

In order to verify the detection effect of the proposed model in edge computing devices, the LubanCat-5 development board (specifications are shown in Table 12) was selected for model deployment. LubanCat-5 is a low-power, high-performance single-board computer designed by Wildfire Technology based on RockChip RK3588. The built-in NPU provides 6TOPS computing power, with extremely low power consumption, only about 1% of GPUs with the same computing power. It supports mainstream deep learning frameworks such as Tensorflow, Pytorch, Caffe, and ONNX, and can realize local offline AI computing in complex industrial scenarios.

The model in this paper is a PT-format model based on the PyTorch framework. Since the neural network accelerator NPU built into the RK3588 development board only supports accelerated inference for RKNN-format models, and model deployment needs to use the NPU for inference acceleration, it is necessary to convert the model format to RKNN. Considering that the RK platform cannot fully support all operators in PyTorch, direct conversion is difficult, while ONNX

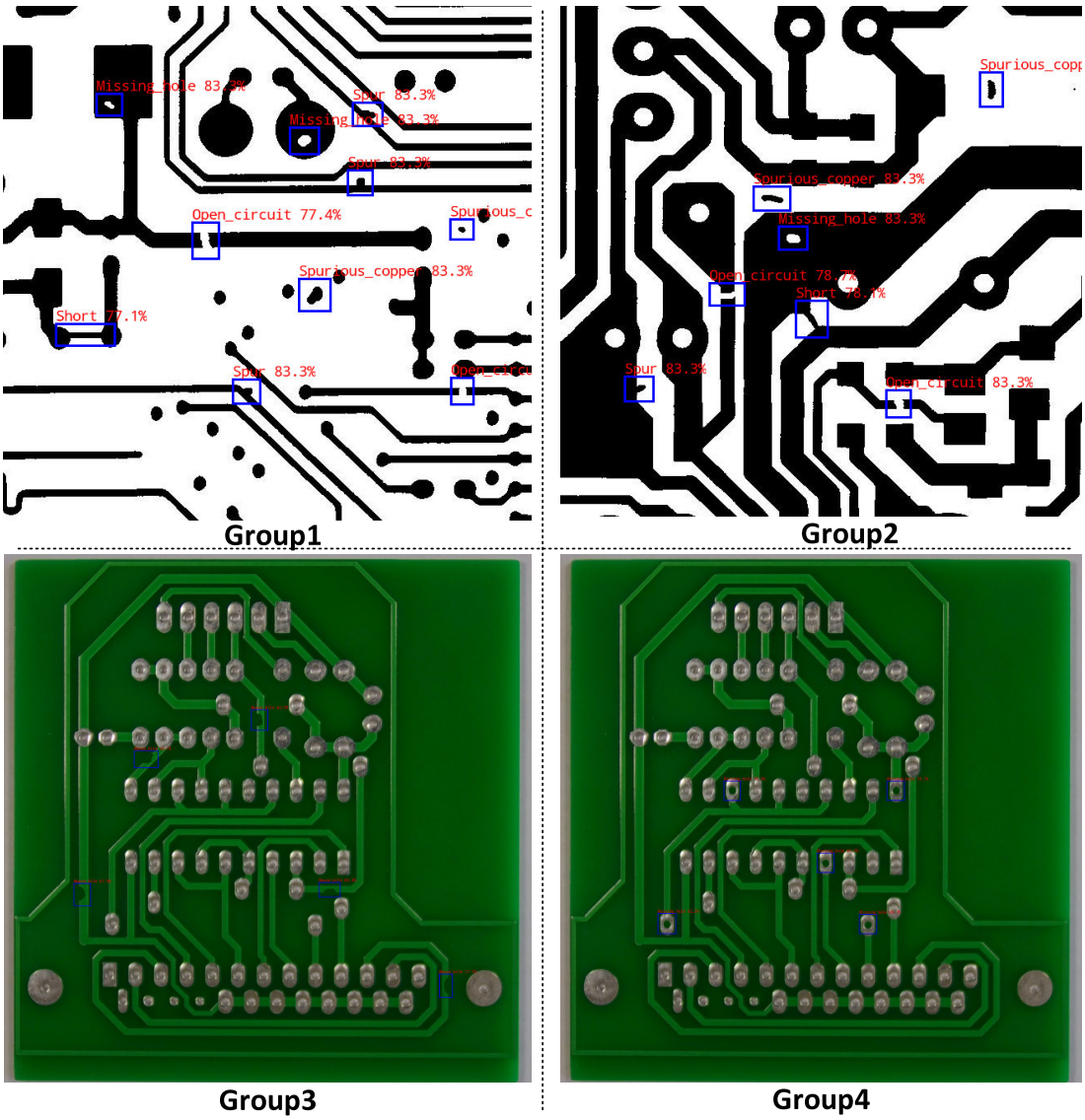


FIGURE 11. Detection effect diagram on the RK3588 platform.

models have good compatibility across various hardware and inference engines. Therefore, this paper adopts an indirect model conversion scheme: PyTorch → ONNX → RKNN. During the model format conversion process, because different formats differ in the encoding, compression, and storage methods of model data, the model size will change accordingly. The sizes of the model before and after improvement are shown in Table 13.

Based on PCB images in the test set, a systematic evaluation of detection performance on the RK3588 edge computing platform was conducted. During the experiments, this paper recorded the time period from image data input to detection result output, and performed statistical analysis with frame rate as the core performance indicator. The video detection process is shown in Fig. 10, intuitively demonstrating the actual running effect of the model on the RK3588 platform. During detection, the detection boxes and confidence are marked in real time, accompanied by detection status and frame rate display. The detection effect of the proposed model on the RK3588 platform is shown in Fig. 11.

In order to visually show the quantization effects of different quantization algorithms, after converting both the proposed model and the original model into RKNN models, comparative experiments were carried out using three schemes: linear quantization [32], asymmetric quantization [33], and no quantization. When inferring the same PCB dataset, the quantization and evaluation results are shown in Table 14.

IV. CONCLUSION

In the process of using deep learning methods for PCB defect detection, it is found that a large number of convolution layers

in deep networks will bring a large amount of computation, and at the same time it is necessary to realize lightweight deployment of the model while ensuring that detection accuracy does not drop. In terms of loss function design, there is still room to improve the accuracy of the bounding regression function. To address the above problems, this paper proposes a lightweight PCB defect detection method based on an improved YOLOv11s. First, the backbone trunk is replaced with a lightweight and efficient EfficientViT module, simplifying the model structure and reducing the number of model parameters. Second, the C3k2_Faster_EMA module is designed to further reduce the number of model parameters and enhance the feature extraction capability of the neck network. Finally, the original bounding loss function CIoU is replaced, integrating the auxiliary bounding-box loss function Inner_IoU with Wise_IoU to improve localization accuracy for tiny targets and enhance bounding-box regression performance. On the PKU-Market-PCB and DeepPCB datasets, compared with the original YOLOv11s algorithm, the proposed algorithm improves mAP by 0.7% to 92.3%, reduces computational cost by 63.89%, reduces parameter count by 61.70%, and reduces model size by 58.33% to 8MB, achieving a network design with high accuracy and lightweighting. In addition, deployment experiments on the RK3588 edge device show that after asymmetric quantization, model performance is significantly improved, meeting the requirements of on-device deployment. In the future, we will further reduce model complexity and improve detection accuracy, in order to realize the application of the proposed algorithm in more industrial vision tasks.

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